Features

- 13.56 MHz ± 7 kHz RFID IC for Cards and Tags
- 320 Read/Write EEPROM Bits, Divided into 10 Pages of 32 Bits
- Supports ISO/IEC 14443-2 for Type A and Type B Systems
- Optional 2-byte CRC
- Password and Write-lock Protection
- Programmable Send and Receive Protocols
- Supports Multiple Tags (Anti-collision)
- Integrated up to 70 pF Tuning Capacitor (mask selectable)
- ID Length Programmable from 4 to 19 Bytes
- Optional Stop and Start Bits
- Programmable for 106K or 10.6K Bits/sec

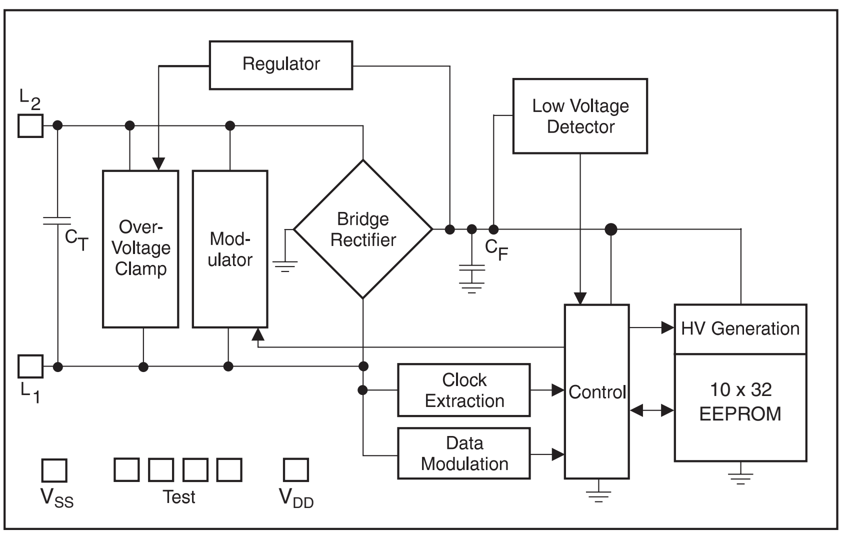
13.56 MHz RFID Transponder IC

AT88RF256-13

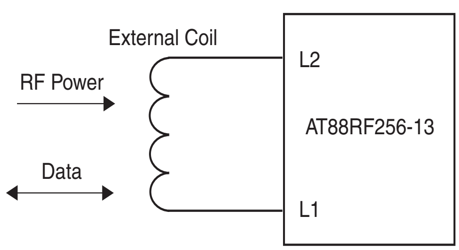
Description

The device is intended to be used in identification applications where one or more ID tags will be in the field at one time. It contains 320 bits of full read/write EEPROM memory, and offers features such as passwords, locking and a variable length ID. It supports all of ISO/IEC14443-2 for both Type A and Type B systems and includes an option for communications at either 106K or 10.6Kbps. The IC includes an internal tuning capacitor – only an external coil antenna is required to form a complete tag.

Block Diagram



Typical Operating Configuration



Rev. 1944A–01/01

IC Operation

Upon power-up, the IC will continuously repeat through the following sequence, which includes an ID transmission and possible reception of a command. The sequence is defined as follows:

1. Framed transmission of the ID field:
   - -Start of Transmission (see *Data Communications* section)
   - -Between 4 and 19 bytes from the EEPROM, which is defined as the ID field
   - -Optional 2-byte CRC
   - -End of Transmission (see *Data Communications* section)
2. A listening window, during which commands may be sent to the IC

All bits are sent to or read from the IC least significant bit first. Bit fields listed in this document are listed with the LSB on the left and the MSB on the right.

Multi-byte information is sent to the IC least significant byte first. Although not visible to the external system, within the IC the first byte sent to the IC is stored in memory at the lowest address and the address is incremented for subsequent bytes.

Information is read from the EEPROM and transmitted by the IC in exactly the same order in which it was written; the first bit written is the first bit read.

ID Field

The ID sent by the IC can be between 4 and 19 bytes in length, depending on the value of the PU_LEN field in the configuration page. EEPROM bytes not utilized for ID storage may be used by the system for any other purpose.

A serial number is programmed into page 7 of the IC at the Atmel factory. This ID is guaranteed to be unique for all dies, but is not locked and can be changed by the system.

CRC

When enabled by the CRC_ON option bit, a 2-byte CRC code will end all frame transmissions (either from the IC or from the reader/writer) under Type A and Type B operation. The CRC polynomial used in this IC is identical to both CRC_A and CRC_B as defined in ISO/IEC 14443-3: $x^{16} + x^{12} + x^5 + x^0$, or a hex polynomial of 1021.

For CRC_A, the CRC register is initialized to 0x6363, while for CRC_B it is initialized to 0xFFFF. When receiving information from the system, the IC computes the CRC on the incoming command, data and CRC bytes (start/stop bits, parity bits, SOT, EOT and EGT are ignored). When the last bit of the CRC has been received, the value in the CRC register should be 0. When the IC transmits data, the CRC is computed based on the data bits. The CRC is transmitted as the final 2 bytes of the frame.

Listening Window

After the power-up sequence is transmitted, there is a listening window during which the tag looks for modulation that would initiate the transmission of a command from the reader/writer to the tag. Commands sent at any other time are ignored.

The listening window is 8 bit-times long. The leading modulation edge of the SOT identifier (see *Data Communications* section) must not start within the first and/or last bit-time of the listening window. This restriction is enforced to prevent the IC receiver from seeing its own modulation.

Command Bytes

The explicit commands implemented in this tag permit the reader/writer to directly access individual 4-byte pages within the memory array, prevent future writing of particular pages (locking), temporarily disable the IC or check a password value. These commands are encoded as follows:

LSB	MSB
A_0 A_1 A_2 A_3 0 0 0 1	Read 32-bit Page A A A A (followed by optional CRC)
A_0 A_1 A_2 0 0 0 1 0	Write 32-bit Data Page A A A (followed by 4 bytes of data and optional CRC)
0 0 0 0 0 0 1 1	Write Lock Byte (followed by 1 byte of data, 3 bytes of $AAh and optional CRC)
0 0 0 1 0 0 1 1	Write Configuration Bits (followed by 1 byte of $AAh, 3 bytes of data and optional CRC)
0 0 0 0 0 1 1 1	Write Password (followed by 4 bytes of data and optional CRC)
0 0 0 1 1 0 0 0	Disable (Stop) IC until Power-down (followed by optional CRC)
0 0 0 1 1 1 0 0	Check Password (followed by 4 bytes of data and optional CRC)

For the Read and all four Write commands, the data stored within the corresponding page of the EEPROM to the accessed page is repeatedly transmitted back to the reader by the IC after the command has completed. This permits a verify function for the commands. For the Write Lock and Write Config commands, the entire contents of page 8 are transmitted. Between each frame transmitted, there is a listening window of 8 bit-times to synchronize the reader and/or permit the reader/writer to issue a new command to the IC. The listening window will begin immediately following the transmission of the appropriate EOT (see *Data Communications* section for information regarding EOT).

For the Read command, addresses 0–7 are used to select between the 32 data pages while an address value of 0x08 addresses the configuration page. Any attempts to read pages with addresses above 0x08 will cause the command to be aborted, and the IC will return to the header transmission sequence.

After the Check Password command, the IC goes back to the ID transmission loop and the reader/writer can issue its commands during the first listening window. After the Disable command, the IC is held in reset until power is removed.

There are a number of features that are used to prevent inadvertent writing of the IC:

1. The proper command code plus the proper receive data encoding must be sent to the IC. If either an illegal code or improper encoding is detected, the command is aborted.
2. Optionally, 2 CRC bytes may be sent after the command and data bytes (see below for details), which must also be correct.
3. In Type A operation, there is a single parity bit following each of the command, data and (optionally) CRC bytes.
4. For both the Write Lock and Write Config commands, some of the bytes in the 32-bit block must have the value AA (hex). If any bit is improperly received, the command is aborted.
5. For the Write Lock command, a successful Write Page command must have been previously executed since the last power cycle in order for the Write Lock command to be executed.

If any of these protections are violated, or if there is a transmission or protection failure (lock bit set, password not entered), or if an illegal command is sent, the part will immediately restart its power-up read sequence.

Passwords

If the optional password mode is enabled with PW_ON, command-based Reads and Writes are prohibited until the correct password is sent using the Check Password command. If the transmitted value of the password is correct, then an internal latch is set and subsequent Read, Write and Lock commands (to any page, including the password page, #9) are permitted. If the wrong password is sent (to the password check), then the command is aborted and the IC reverts to the normal power-up sequence. Writes to locked pages are never permitted regardless of passwords.

There is no command that can be used to directly read the password page, regardless of whether or not the password option (PW_ON) is enabled.

Data Locking

Within the lock byte, each lock bit determines whether the corresponding 4-byte user page can be written to. If it is a "1", then Writes are prohibited; if "0", they are allowed. The data sent to the IC with the Write Lock operation is OR'ed with the data already in the lock byte and then rewritten to the EEPROM. Once a user page is locked, it may never be unlocked and may never be written to.

There are two additional lock bits for pages 8 (CONFIG_LOCK) and 9 (PW_LOCK). They operate slightly differently from the user lock bits because there is no OR function. CONFIG_LOCK, if "1", prevents the execution of the Write Config Bits command, while PW_LOCK, if "1", prevents execution of the Write Password command. Turning on CONFIG_LOCK does not lock the value of the bits within the lock byte but does prevent further change to the PW_LOCK bit.

Multiple Tags

In order to support multiple tags within the field at the same time, a random delay time between ID transmissions can be enabled. This feature is implemented by having the IC randomly disable its activity (transmission of ID frame *and* enabling the listening window) at selected times. Commands are only honored during a listening window immediately following a frame transmission.

The IC includes a random generator that will generate different sequences of enablement/disablement based on processing, voltage, temperature and power-up time. Depending on the value of the RANDOM option, the transmission of an ID frame will be enabled on average once in 8, 32 or 128 times. The maximum delay is twice the average, while at the minimum, two ID frames/listening windows may be issued back-to-back.

To implement this feature, the tags must be programmed with error detection information within the ID field so that the reader can detect the condition when two tags transmit their ID at exactly the same time. Because of the random delay feature, in most cases the next transmissions for these two ICs will not overlap.

The Chip Disable command can be used with the random delay feature to permit an increased number of tags to be identified. Once a tag has been properly read by the reader unit, the reader sends the Chip Disable command to the tag during the first listening window after the ID transmission. Until the power is removed, that tag no longer sends its ID frame.

Data Communications

The electrical signaling of the IC is configurable using the TYPE_14443, DATA_RATE[0:1] and DATA_ENCODE bits in the configuration page. Options exist to allow compatibility with ISO/IEC 14443-2 "Radio frequency power and signal interface" (version N409 Final Committee Draft 3/12/99) Type A and Type B modes. The frame formatting for the various modes of operation is not fully compliant with ISO/IEC 14443 and is defined below.

Electrical Signaling

The electrical signaling between the reader/writer and the IC is configured using the TYPE_14443, DATA_RATE[0:1] and DATA_ENCODE bits of the configuration page. These options are described as follows:

- *TYPE_14443* – This bit selects either ISO/IEC 14443-2 Type A or Type B electrical signaling for incoming data (reader/writer to the IC). A “1” on this bit selects Type B operation and a “0” selects Type A operation.
- *DATA_RATE[0:1]* – These bits select the data rate for incoming (reader/writer to IC) and outgoing (IC to reader/writer) communications. For outgoing communication, these bits also select the PSK subcarrier rate. Selection “0x” is compliant with ISO/IEC 14443-2. See *Option Page* section on page 12 for more information.
- *DATA_ENCODE* – This bit determines the outgoing (IC to reader/writer) data encoding format.

When the TYPE_14443 option bit is “1” (Type B mode):

0 - selects BPSK-NRZ-L, which is compliant with ISO/IEC 14443-2 Type B. When BPSK-NRZ-L is selected, the subcarrier is modulated with the carrier. A 180° phase change of the subcarrier occurs whenever the data state changes.

1 - selects BPSK-Miller encoding. When BPSK-Miller is selected, the subcarrier is modulated with the carrier. For a logic “1”, a 180° phase transition of the subcarrier will occur in the middle of the bit-time. For a logic “0” following a logic “1”, there will be no phase change of the subcarrier during the bit-time. For a logic “0” following a logic “0”, there will be a 180° phase change of the subcarrier at the beginning of the bit-time.

When the TYPE_14443 option bit is “0” (Type A mode):

0 - selects the subcarrier gated with Manchester, which is compliant with ISO/IEC 14443-2 Type A. When this mode is selected, the data bit is Manchester encoded during each bit-time. The Manchester-encoded data is then logically ANDed with the subcarrier. If the subcarrier and the Manchester-encoded data are both a “1”, then the field will be modulated; otherwise, the field will not be modulated.

A Manchester logic “1” is encoded as a “1” for the first half of the bit-time and a “0” for the second half of the bit-time. A Manchester “0” is encoded as a “0” for the first half of the bit-time and a “1” for the second half of the bit-time.

1 - selects subcarrier gated with Miller. When this mode is selected, the data bit is Miller encoded during each bit-time. The Miller-encoded data is then logically ANDed with the subcarrier. If the subcarrier and the Miller-encoded data are both a “1”, then the field will be modulated; otherwise, the field will not be modulated.

A Miller logic “1” transitions from its current state to the opposite state in the middle of the bit-time. A Miller logic “0” following a logic “1” will remain at its previous state for the entire bit time. A Miller logic “0” following a logic “0” will transition from its current state to the opposite state at the beginning of the bit-time. The initial Miller state is “0” when the IC begins transmitting its SOT.

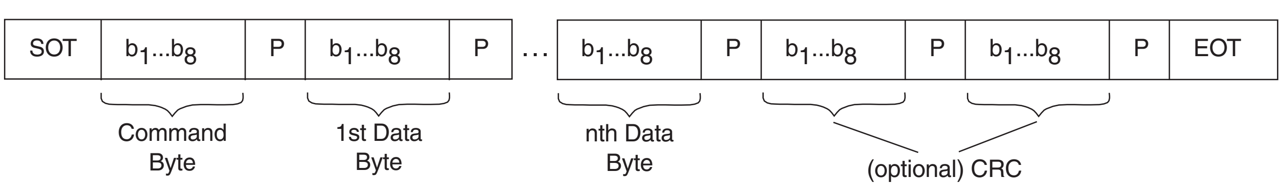
Type A Frame Mode

Setting the TYPE_14443 option bit to “0” puts the IC in Type A mode. The following Type A information applies only when TYPE_14443 is “0”.

All commands and data sent to the IC in Type A mode shall be received in the following frame format:

1. Start of Transmission (SOT) (see below for Type A SOT definition)
2. Eight command bits + 1 odd parity bit
3. N*(8 data bits + 1 odd parity bit) $N \geq 0$
4. (Optional) Two additional CRC bytes in the same format as (3) above
5. End of Transmission (EOT) – see below for Type A EOT definition

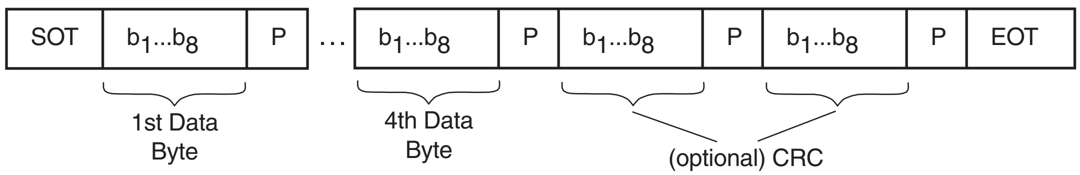
Figure 1. Type A Reader/Writer to IC Frame Format



All data sent from the device to the reader/writer will be transmitted in the following frame format:

1. Start of Transmission (SOT)
2. N*(8 data bits + 1 odd parity bit) N ≥ 4
3. (Optional) Two additional CRC bytes in the same format as (2) above
4. End of Transmission (EOT)

Figure 2. Type A IC to Reader/Writer Frame Format



Type A Definitions

SOT/EOT	Definition
Start of Transmission (reader/writer to IC)	The SOT received by the IC is defined as bit-duration containing a “pause” (or 100% modulation) at the beginning of the bit-duration. For more information, see *Start of Communication* in ISO/IEC 14443-2, section 8.1.3, which is identical to SOT as referred to in this document.
End of Transmission (reader/writer to IC)	The EOT received by the IC is defined as a logic “0” followed by 1 bit-duration with no carrier modulation. For more information, see “end of communication” in ISO/IEC 14443-2, section 8.1.3, which is identical to EOT as referred to in this document.
Start of Transmission (IC to reader/writer)	The SOT transmitted by the IC to the reader/writer will be a logic “1” and will be encoded according to the DATA_ENCODE selection made in the option page. See Figure 3.
End of Transmission (IC to reader/writer)	The EOT transmitted by the IC to the reader/writer will be a logic “0” and will be encoded according to the DATA_ENCODE selection made in the option page. See Figure 4. Note that there are four possibilities for this if Miller is selected.

Figure 3. Type A SOT (IC to Reader)

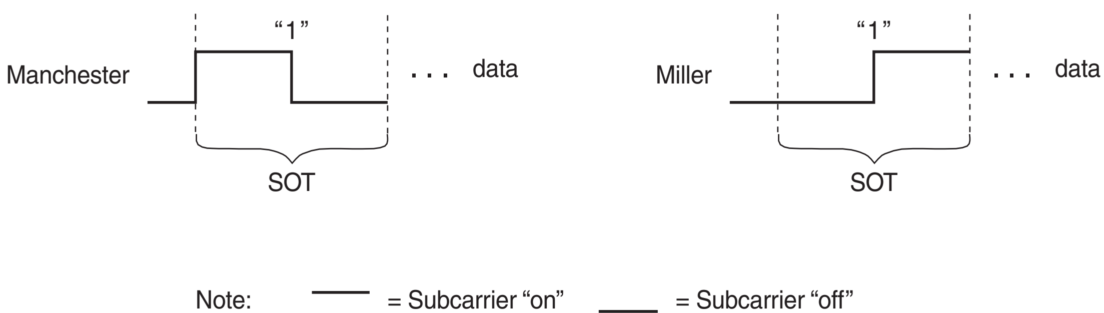
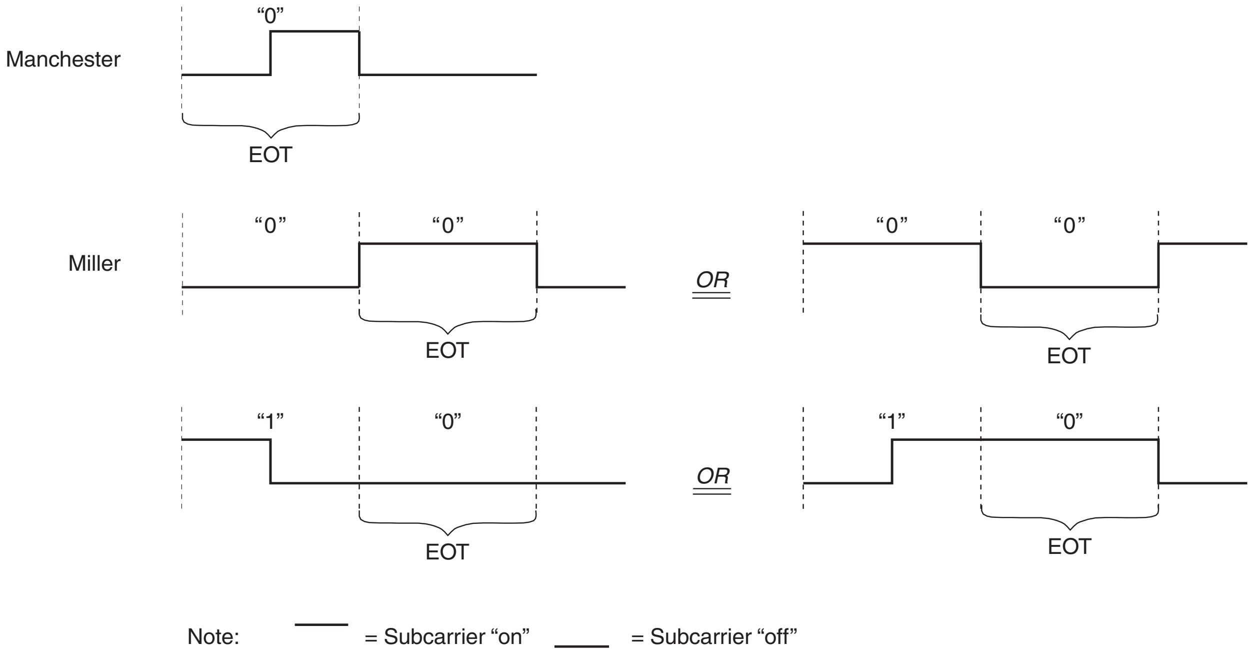


Figure 4. Type A EOT (IC to Reader)



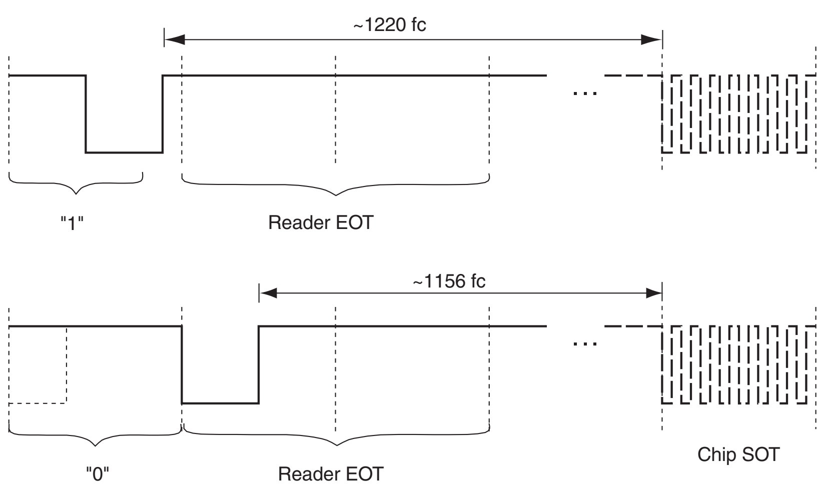
Type A Timing

The IC will respond to a reader/writer Read Page or Check Password command according to the timing indicated in Figure 5. The response to a successful Read Page command will be the requested page data. The response to the Check Password command will be a new framed transmission of the ID field (see page 1).

If there is an error of one sort or another with either of these commands, the IC will also start a new framed transmission of the ID field.

Figure 5 refers to an IC configured to transmit at the ISO standard rate of 106 Kbps using the standard Manchester encoding. If the 10.6 Kbps data rate is chosen, then the delays will be 12164 and 11524 cycles, respectively. If Miller encoding is chosen, then there will be an additional one-half bit-time before modulation starts on the SOT.

Figure 5. Type A Frame Timing Information (ISO 14443 Timing and Encoding)



Type B Frame Mode

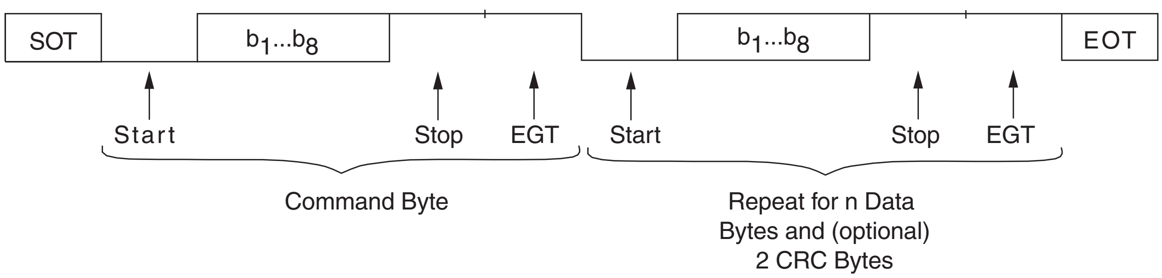
Setting the TYPE_14443 option bit to “1” puts the IC in Type B mode. The following Type B information applies only when TYPE_14443 is “1”.

All commands and data sent to the IC in Type B mode shall be received in the following format:

1. Start of Transmission (SOT) – see below for Type B SOT definition)
2. One command byte (1 logic “0” start bit + 8 command bits + 1 logic “1” stop bit + EGT(1))
3. N data bytes - N*(1 logic “0” start bit + 8 data bits + 1 logic “1” stop bit + EGT(1)) $N \geq 0$
4. (Optional) Two CRC bytes in the same format as (3) above
5. End of Transmission (EOT) – see below for Type B EOT definition

Note: EGT (Extra Guard Time) is 0–6 additional logic “1”s transmitted after stop bit.

Figure 6. Type B Frame Format (Reader/Writer to Card)

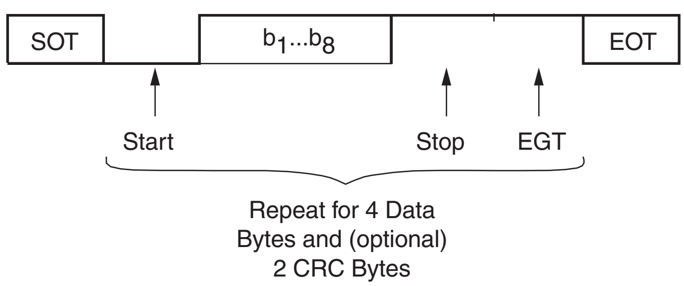


All data sent from the IC to the reader/writer will be transmitted in the following format:

1. Start of Transmission (SOT)
2. N data bytes - N (1 logic “0” start bit + 8 data bits + 1 logic “1” stop bit + 1 EGT(1) bit) $N \geq 4$
3. (Optional) Two CRC bytes in the same format as (2) above
4. End of Transmission (EOT)

Note: The EGT (Extra Guard Time) bit transmitted from the IC to the reader/writer is a logic “1”.

Figure 7. Type B Frame Format (IC to Reader/Writer)



Type B SOT/EOT

The Start of Transmission and End of Transmission sequences for Type B operation are defined below:

Start of Transmission (reader/writer to IC):

1. No carrier modulation.
2. The carrier is modulated (logic "0") for 10 or 11 bit-times (integer multiples).
3. The carrier is not modulated (logic "1") for 2 or 3 bit-times (integer multiples).

End of Transmission (reader/writer to IC):

1. No carrier modulation (this is the stop bit or EGT bit of the last data/command byte).
2. The carrier is modulated (logic "0") for 10 or 11 bit-times.
3. The reader stops modulating the carrier (logic "1").

Start of Transmission (IC to reader/writer):

1. The carrier is modulated with the subcarrier for 8 bit-times.
2. The subcarrier is phase shifted 180°.
3. The subcarrier is modulated with the carrier for 10 bit-times with no subcarrier phase change.
4. The subcarrier is phase shifted 180°.
5. The subcarrier is modulated with the carrier for 2 bit-times with no subcarrier phase changes.

Note: There will be a 180° phase change of the subcarrier between the end of the SOT and the start bit "0" if the DATA_ENCODE option is set to BPSK-NRZ-L. There will be no phase change of the subcarrier between the end of the SOT and the start bit "0" if the DATA_ENCODE option is set to BPSK-Miller.) See Figure 8 below.

End of Transmission (IC to reader/writer). See Figure 9 on page 10:

1. A logic "0" (encoded in the format selected by the DATA_ENCODE option bit) is transmitted for 10 bit-times.
2. A logic "1" (encoded in the format selected by the DATA_ENCODE option bit) is transmitted for 2 bit-times.
3. The IC stops the subcarrier modulation.

Figure 8. Type B SOT (IC to Reader)

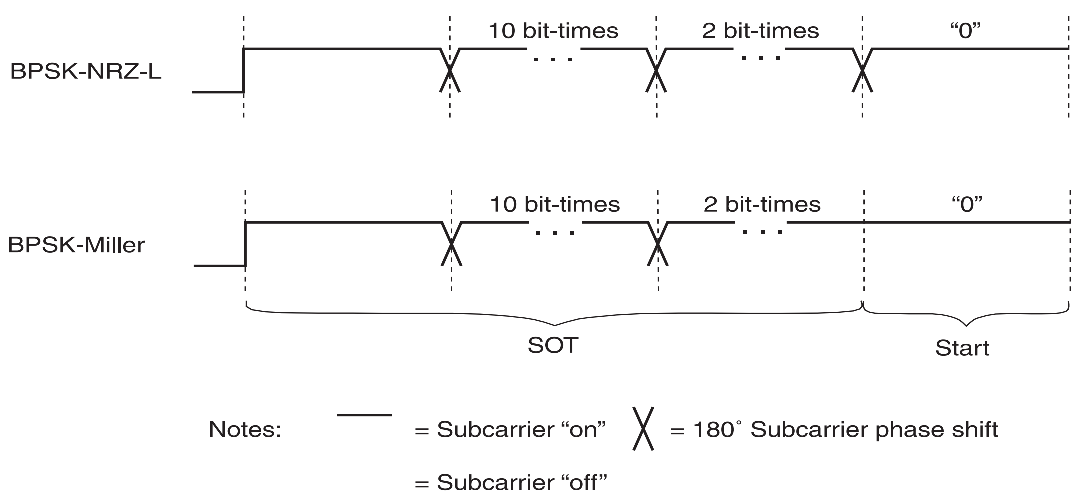
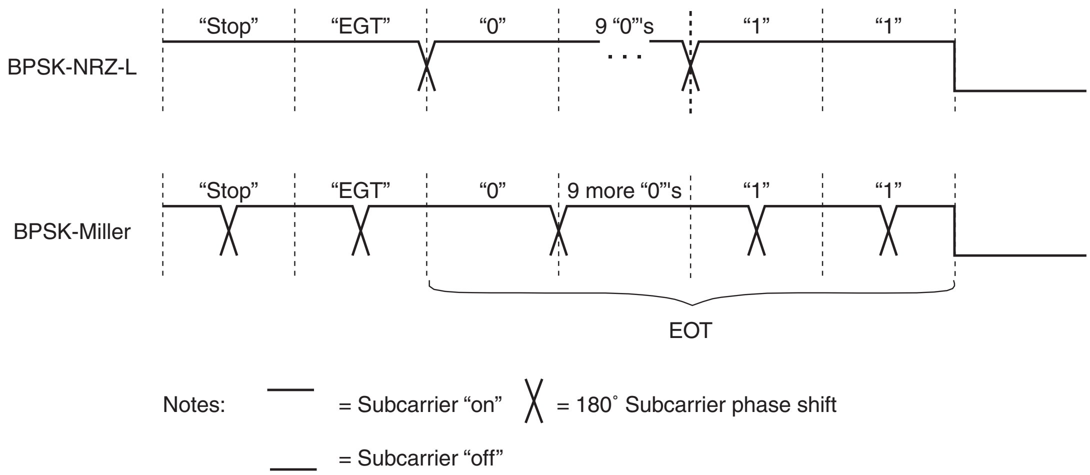


Figure 9. Type B EOT (IC to Reader)

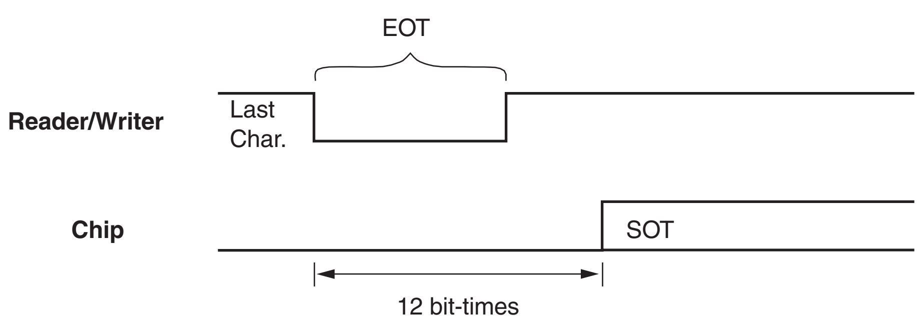


Type B Timing

The IC will begin transmitting the SOT sequence (in response to the reader/writer's Read Page or Check Password command frame) exactly 12 bit-times from the beginning of the reader/writer's EOT. The response to a successful Read Page command will be the requested page data. The response to the Check Password command will be a new framed transmission of the ID field (see page 1).

If there is an error of one sort or another with either of these commands, the IC will also start a new framed transmission of the ID field.

Figure 10. Type B Frame Timing Restrictions



Reset Voltage

The device includes a precision voltage reference to ensure that all Write commands are only performed when the power-supply voltage on the IC is above a required level of 2.0V. ID Reads and the Read and Disable commands will take place regardless of voltage (above a minimal POR level), which will result in correct information in most cases. Data transmitted at the lowest voltages may not be valid, and therefore some sort of error detection and/or correction (multiple reads, parity, hamming code, etc.) must be implemented by the system.

Mechanical Specification

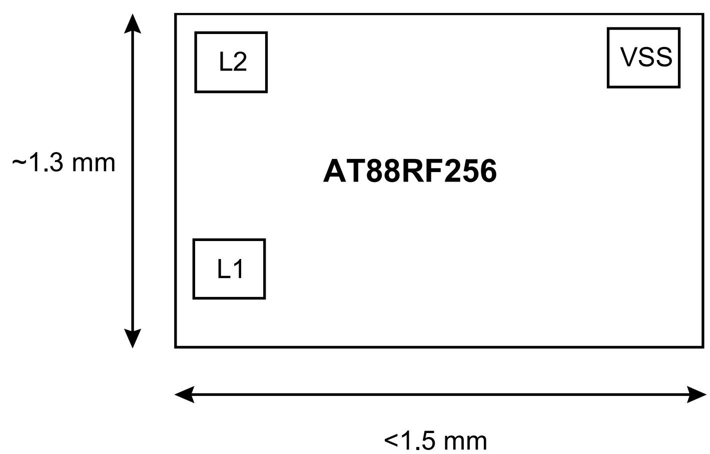
The IC will contain two coil input pads with ESD protection, at levels greater than 2 kV, along one end of the IC. These two pads will be suitable for wire bonding or solder bumping. All remaining test pads will use a different structure and size, for which production bonding or bumping will not be permitted. ESD protection for these pads is 300V.

The IC includes a tuning capacitor across the coil input pins. Parasitic capacitance across these pins will be less than 10 pF and will vary with voltage, temperature and process.

The IC will be designed in such a way that the EEPROM and analog circuitry can be fully tested via extra test pins. When the test pins are used to access the IC, all protection (passwords, locks, etc.) will be disabled. After testing has been completed, a fuse will be blown from one of the test pads to prevent further use of the test mode features.

A preliminary drawing of the IC is shown below:

Figure 11. AT88RF256-13 Die



Memory Map

The EEPROM is composed of 10 pages of 32 bits each for a total of 320 bits. Pages 0–7 are the user pages, which include ID information and other user-defined bytes. Page 8 is the configuration page, which includes the lock and option bits. Page 9 is the password page.

	Byte 0	Byte 1	Byte 2	Byte 3
Page 0	First ID Byte	Second ID Byte	Third ID Byte	Fourth ID Byte
Page 1	Fifth ID Byte/User Data	Sixth ID Byte/User Data	Seventh ID Byte/User Data	Eighth ID Byte/User Data
Page 2	ID/User Data	ID/User Data	ID/User Data	ID/User Data
Page 3	ID/User Data	ID/User Data	ID/User Data	ID/User Data
Page 4	ID/User Data	ID/User Data	ID/User Data	User Data
Page 5	User Data	User Data	User Data	User Data
Page 6	User Data	User Data	User Data	User Data
Page 7	User Data	User Data	User Data	User Data
Page 8	LOCK0…LOCK7	OPTIONS	OPTIONS	UNUSED
Page 9	First Byte Password	Second Byte Password	Third Byte Password	Fourth Byte Password

Option Page

Bits are listed below in the order in which they must be sent to the IC when the Write Lock Byte or Write Config Bits command is sent to the IC.

When reading this page, all 32 bits are read in this order. This page cannot be written with a single 4-byte frame; the Write Lock Byte command is used for the first byte only, and the Write Config Bits command is used for the last 3 bytes.

The Define column reflects the default value that the options have upon shipment from the Atmel factory.

Lock Byte

Name	Number of Bits	Default Value	Description
LOCK[0:7]	8	0	If 1, locks the corresponding user page against further writes.

Configuration Bits

Name	Number of Bits	Default Value	Description
PU_LEN[0:3]	4	0000	Number of ID bytes after first four. Total ID size range: 4–19 bytes
TEST	1	–	Ignored by IC
RANDOM	1	0	Frames (ID + 8 bit-time listening window) between ID transmissions 0 Continuous frames 1 Random null frames, mean number = 8
PW_ON	1	0	Password enable, if 1 password is page 9 of EEPROM
TYPE_14443	1	1	If 0, use ISO/IEC 14443-2 (Type A); if 1, use ISO/IEC 14443-2 (Type B) for those areas not controlled by the options listed below.
DATA_RATE[0:1]	2	00	Transmit and receive rate. If TYPE_14443 is 1 (Type B), then this field is ignored and the data rate is fixed at 106K bits/sec. 0x 106K bits/sec (ISO/IEC 14443-2 std)—subcarrier is carrier divided by 16 10 10.6K bits/sec—subcarrier is carrier divided by 64 11 10.6K bits/sec—subcarrier is carrier divided by 128
DATA_ENCODE	1	0	Transmit data using the following encoding if TYPE_14443 is “1” 0 BPSK-NRZ-L (ISO/IEC14443-2 Type B standard) 1 BPSK-Miller Transmit data using the following encoding if TYPE_14443 is “0” 0 Subcarrier gated w/Manchester (ISO/IEC 14443-2 Type A standard) 1 Subcarrier gated w/Miller (the initial Miller state will be “1” upon SOT)
PW_LOCK	1	0	If 1, locks the password page against further writes
CONFIG_LOCK	1	0	If 1, locks the configuration page (but not LOCK[7:0]) against further writes
CRC_ON	1	1	If 1, enables 2-byte CRC at the end of each frame transmission
MSB	2	N/A	Don’t Care (ignored by IC)

Engineering Samples

Engineering samples are provided in Atmel’s 14S (14-lead, 0.150” wide, plastic gull wing, small outline [JEDEC SOIC]) package. See *General Information* section at www.atmel.com for the package drawing.

Pin Connections

L1 is connected to pin 10

L2 is connected to pin 11

All the remaining pins should float.

Engineering samples come with a 30 pF tuning capacitor. If no additional capacitance is added (including parasitics), a 4.54 µH nominal antenna inductance should be used. The antenna should be tuned to 13.54 MHz for maximum performance.

Atmel Headquarters

Corporate Headquarters
2325 Orchard Parkway
San Jose, CA 95131
TEL (408) 441-0311
FAX (408) 487-2600

Europe
Atmel SarL
Route des Arsenaux 41
Casa Postale 80
CH-1705 Fribourg
Switzerland
TEL (41) 26-426-5555
FAX (41) 26-426-5500

Asia
Atmel Asia, Ltd.
Room 1219
Chinachem Golden Plaza
77 Mody Road Tsimhatsui
East Kowloon
Hong Kong
TEL (852) 2721-9778
FAX (852) 2722-1369

Japan
Atmel Japan K.K.
9F, Tonetsu Shinkawa Bldg.
1-24-8 Shinkawa
Chuo-ku, Tokyo 104-0033
Japan
TEL (81) 3-3523-3551
FAX (81) 3-3523-7581

Atmel Operations

Atmel Colorado Springs
1150 E. Cheyenne Mtn. Blvd.
Colorado Springs, CO 80906
TEL (719) 576-3300
FAX (719) 540-1759

Atmel Rousset
Zone Industrielle
13106 Rousset Cedex
France
TEL (33) 4-4253-6000
FAX (33) 4-4253-6001

Atmel Smart Card ICs
Scottish Enterprise Technology Park
East Kilbride, Scotland G75 0QR
TEL (44) 1355-803-000
FAX (44) 1355-242-743

Atmel Grenoble
Avenue de Rochepleine
BP 123
38521 Saint-Egreve Cedex
France
TEL (33) 4-7658-3000
FAX (33) 4-7658-3480

Fax-on-Demand
North America:
1-(800) 292-8635
International:
1-(408) 441-0732

e-mail
literature@atmel.com

Web Site
http://www.atmel.com

BBS
1-(408) 436-4309

1944A–01/01/xM